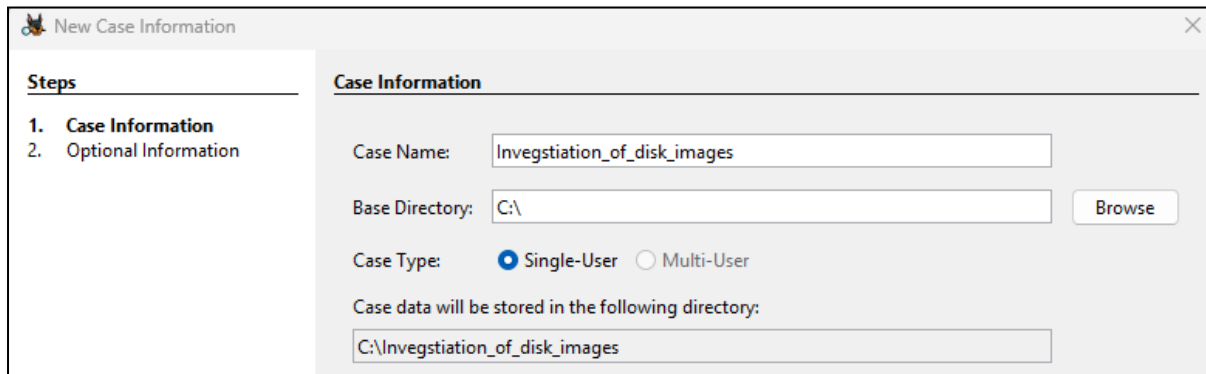


Experiment 05:

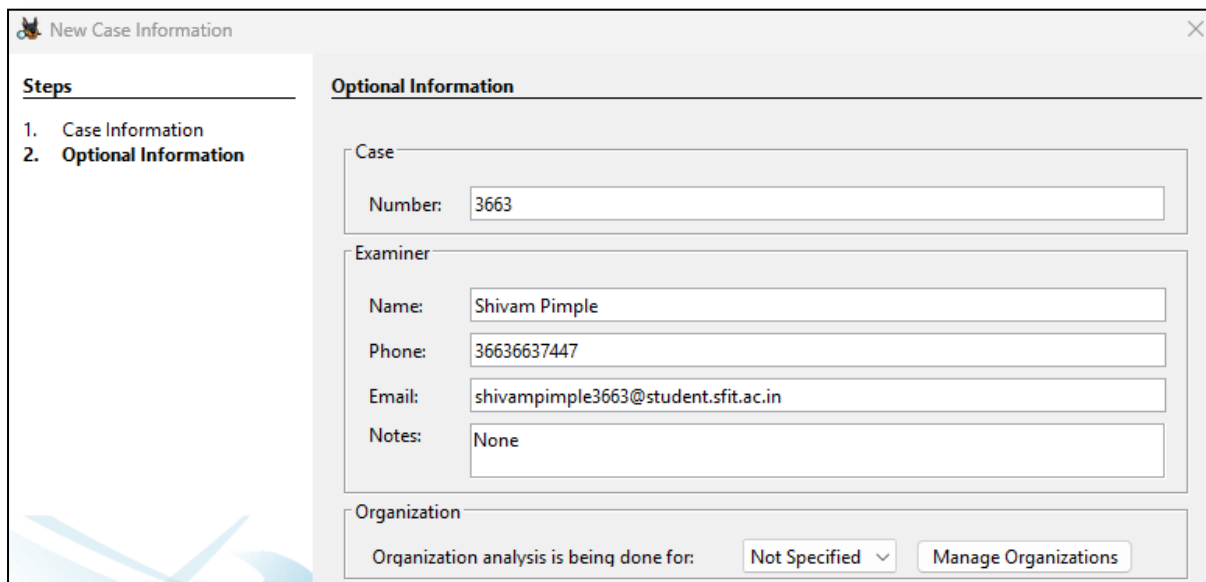
Laboratory Exercise

1. Launching Autopsy and creating a new forensic case.



The screenshot shows the 'New Case Information' dialog box with the 'Case Information' tab selected. The 'Steps' list on the left shows '1. Case Information' and '2. Optional Information'. The 'Case Information' section contains the following fields:

- Case Name:
- Base Directory:
- Case Type: ☒ Single-User ☐ Multi-User
- Case data will be stored in the following directory:



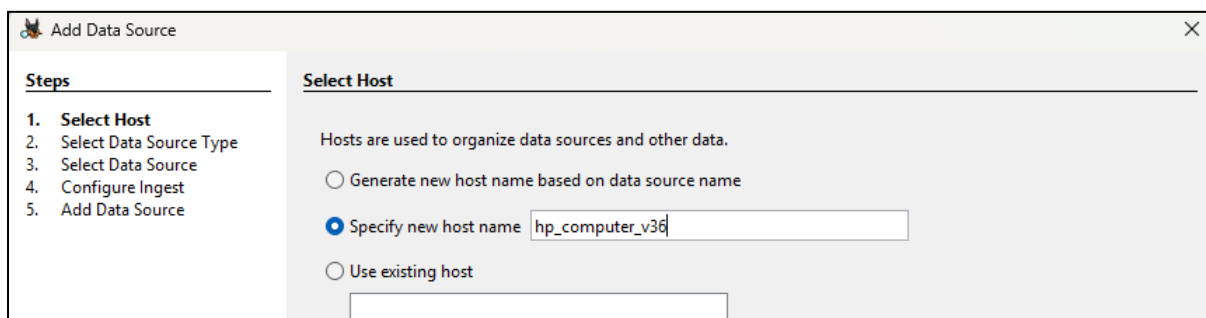
The screenshot shows the 'New Case Information' dialog box with the 'Optional Information' tab selected. The 'Steps' list on the left shows '1. Case Information' and '2. Optional Information'. The 'Optional Information' section contains the following fields:

- Case Number:
- Examiner Name:
- Examiner Phone:
- Examiner Email:
- Examiner Notes:
- Organization analysis is being done for:

Observation:

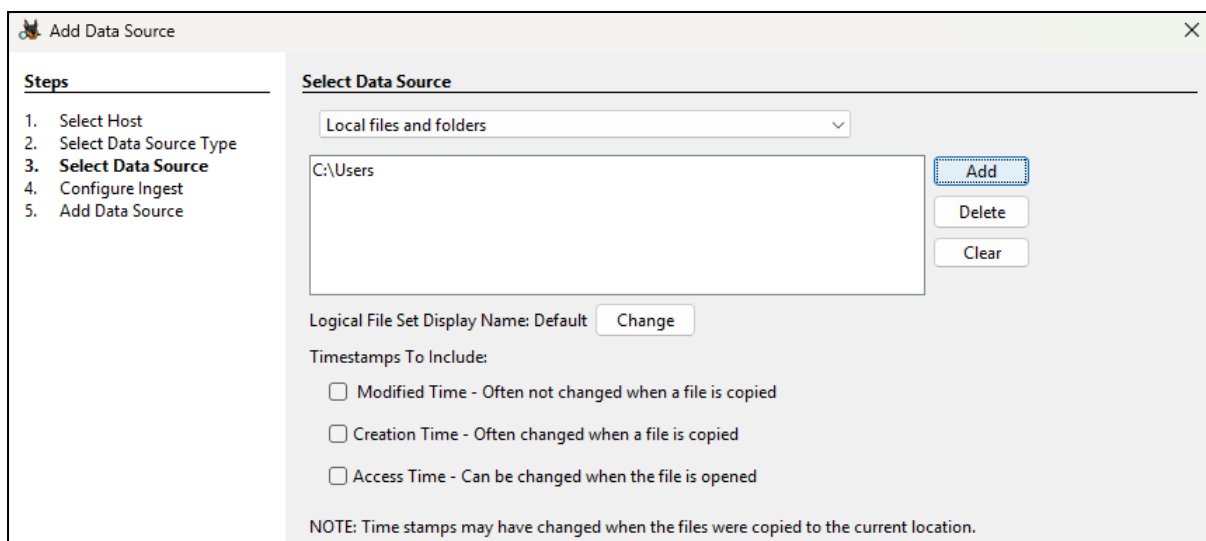
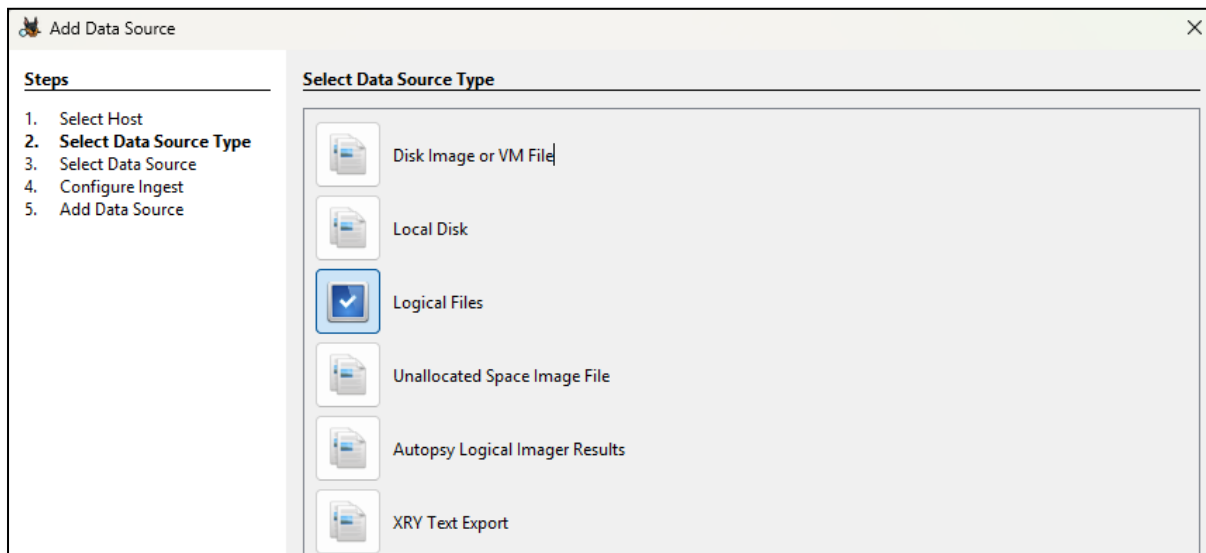
Initialized case **Investigation_of_disk_images** under examiner **Shivam Pimple** with case number **3663**. Configured the case as a **single-user** forensic investigation.

2. Adding a logical image as a data source.



The screenshot shows the 'Add Data Source' dialog box with the 'Select Host' tab selected. The 'Steps' list on the left shows '1. Select Host', '2. Select Data Source Type', '3. Select Data Source', '4. Configure Ingest', and '5. Add Data Source'. The 'Select Host' section contains the following fields:

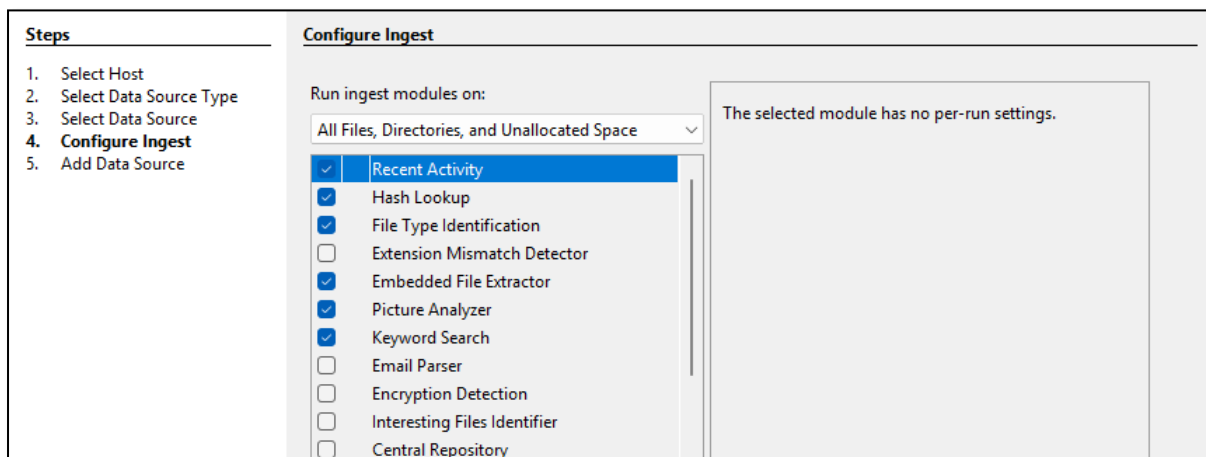
- Hosts are used to organize data sources and other data.
- ☐ Generate new host name based on data source name
- ☒ Specify new host name
- ☐ Use existing host

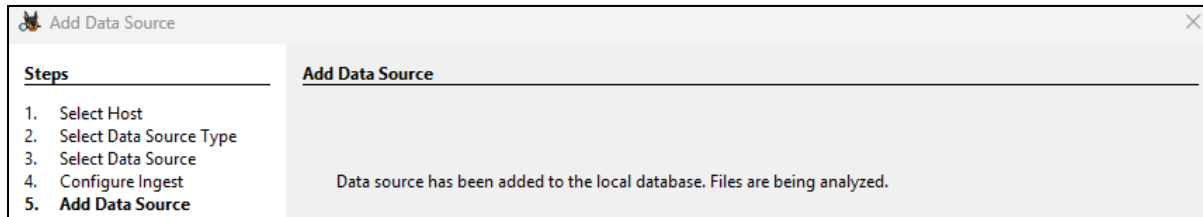
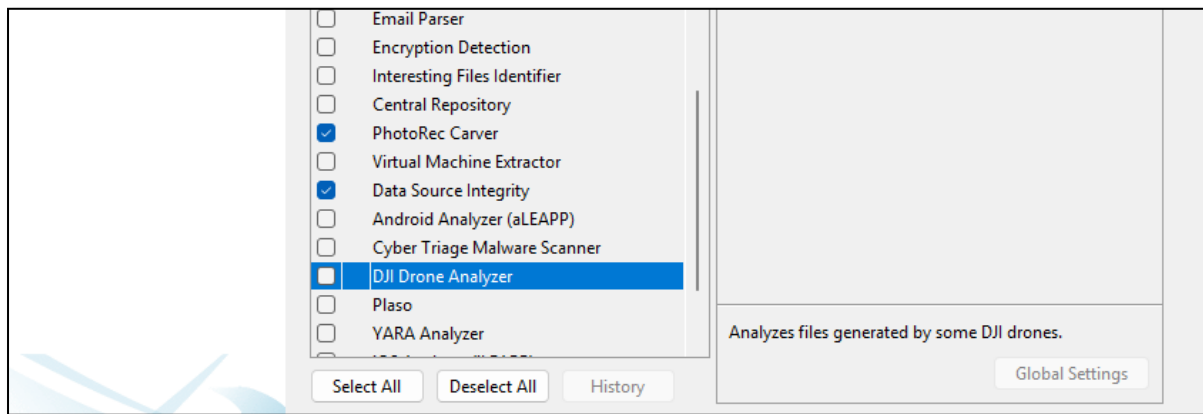


Observation:

Set the host name to **hp_computer_34**, selected **Logical Files**, and targeted the local user directory **C:\Users**. Added the selected path as a data source for ingestion.

3. Configuring ingest modules.

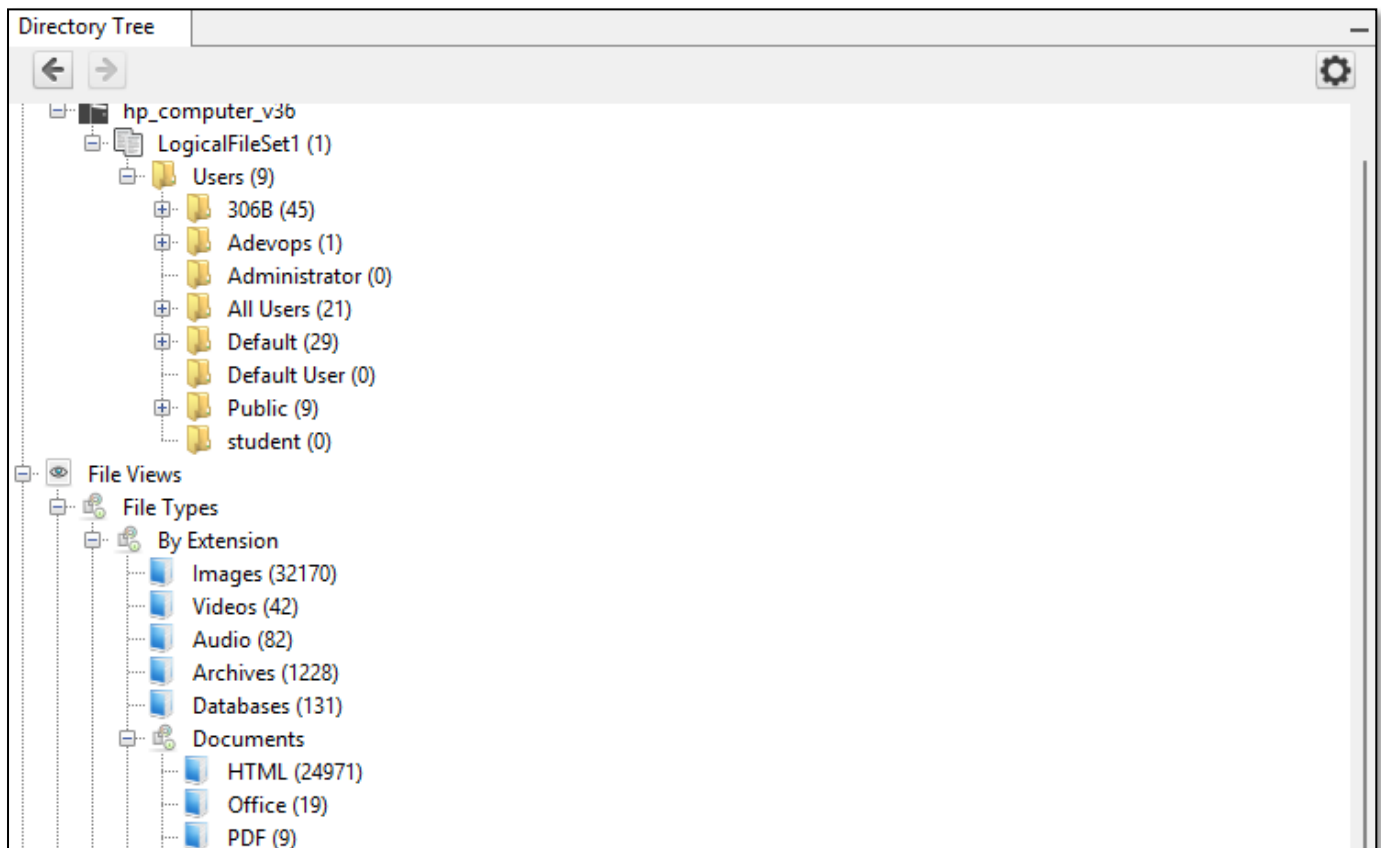




Observation:

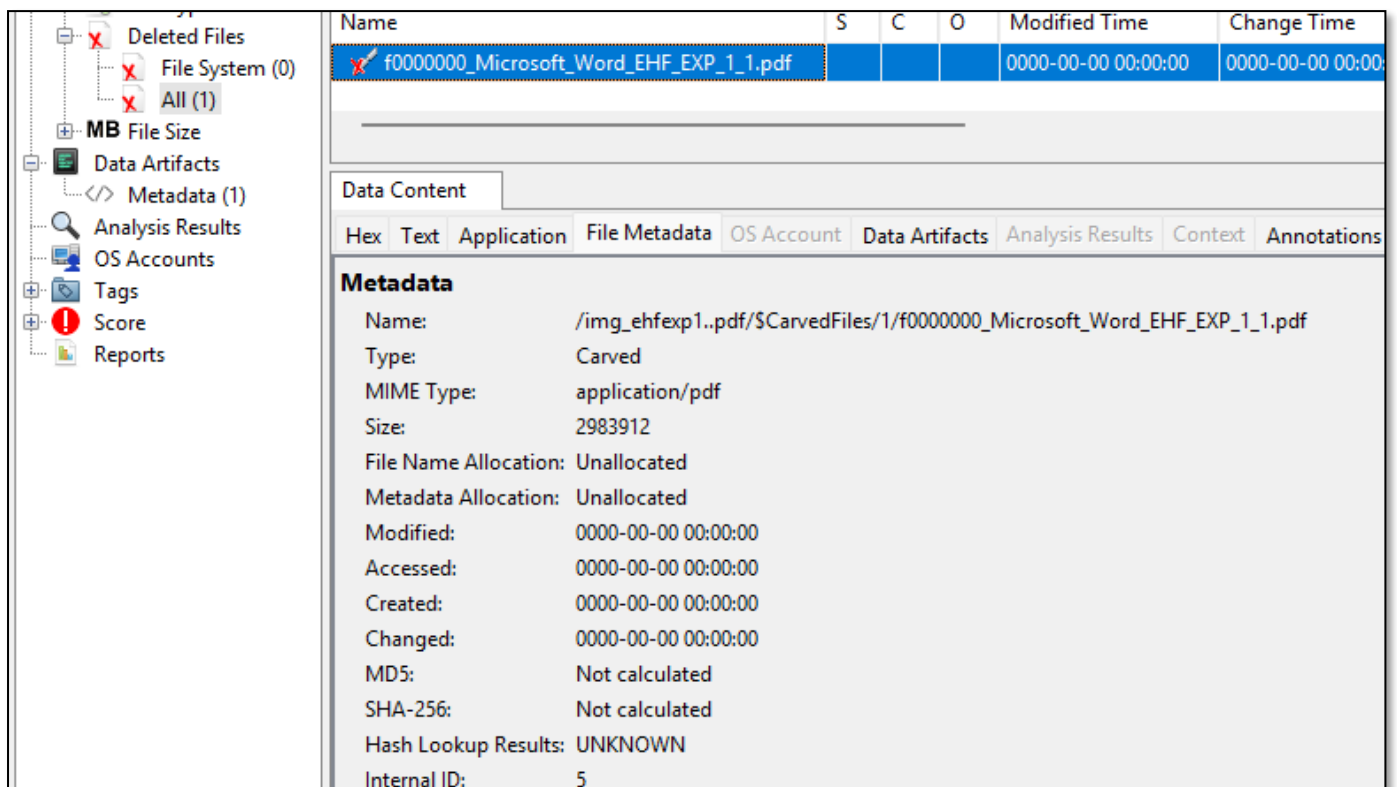
Enabled analysis modules including **Recent Activity**, **Hash Lookup**, **File Type Identification**, **Embedded File Extractor**, **Picture Analyzer**, **Keyword Search**, **PhotoRec Carver**, and **Data Source Integrity** for processing the data source.

4. File system analysis results.



Observation:

Indexed **LogicalFileSet1** and categorized the files into types such as **32,170 images, 42 videos, 62 audio files, 1,228 archives, and 131 databases**. Displayed the internal directory hierarchy of user profile **306B**.

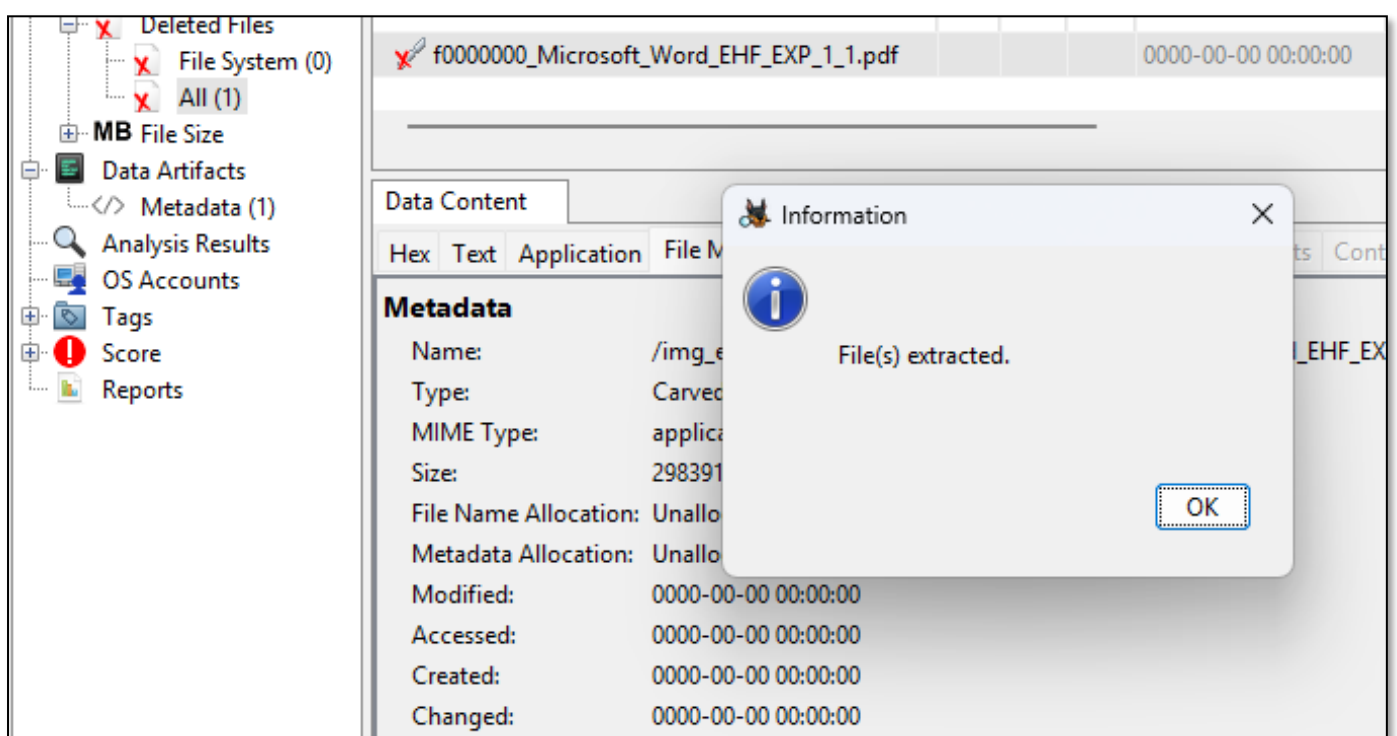
5. Deleted files recovered by Autopsy.


The screenshot shows the Autopsy interface with the 'Deleted Files' section selected in the left sidebar. The main pane displays a table of recovered files. The file 'f0000000_Microsoft_Word_EHF_EXP_1_1.pdf' is highlighted. Below the table, the 'Data Content' tab is active, showing the 'Metadata' for the selected file.

Name	S	C	O	Modified Time	Change Time
f0000000_Microsoft_Word_EHF_EXP_1_1.pdf				0000-00-00 00:00:00	0000-00-00 00:00:00

Metadata

- Name: /img_ehfexp1..pdf/\$CarvedFiles/1/f0000000_Microsoft_Word_EHF_EXP_1_1.pdf
- Type: Carved
- MIME Type: application/pdf
- Size: 2983912
- File Name Allocation: Unallocated
- Metadata Allocation: Unallocated
- Modified: 0000-00-00 00:00:00
- Accessed: 0000-00-00 00:00:00
- Created: 0000-00-00 00:00:00
- Changed: 0000-00-00 00:00:00
- MD5: Not calculated
- SHA-256: Not calculated
- Hash Lookup Results: UNKNOWN
- Internal ID: 5

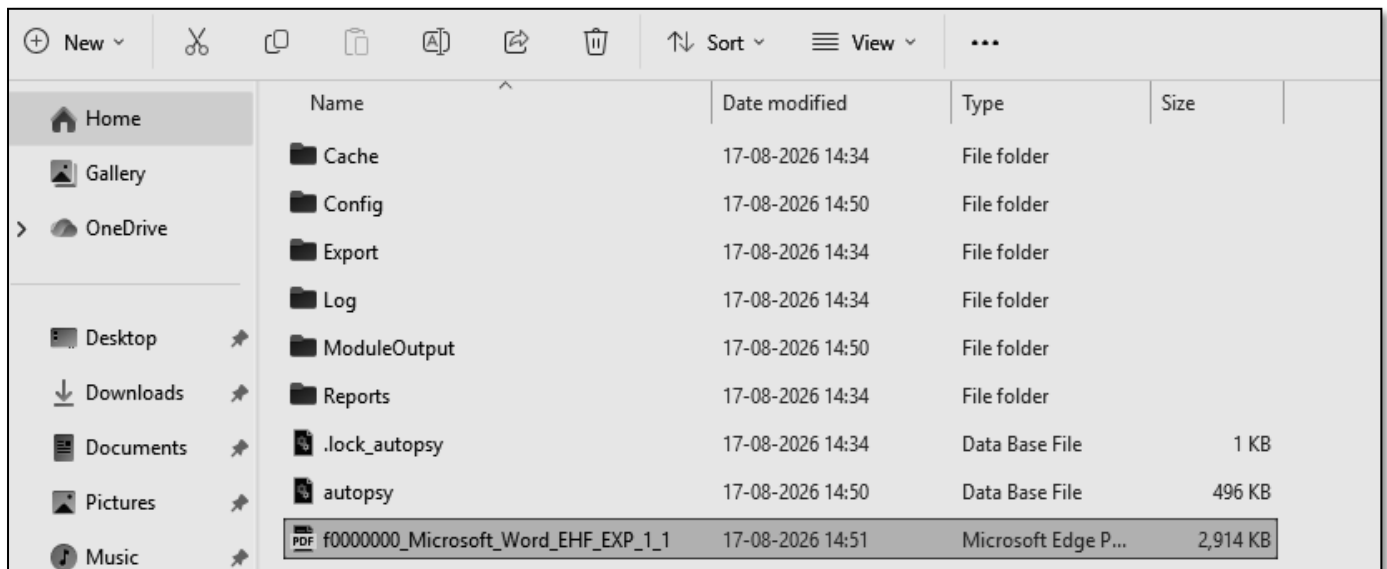


The screenshot shows the Autopsy interface with the 'Deleted Files' section selected. An 'Information' dialog box is displayed over the main pane, indicating that the file(s) have been successfully extracted. The background shows the same file 'f0000000_Microsoft_Word_EHF_EXP_1_1.pdf' and its metadata.

Information

File(s) extracted.

OK

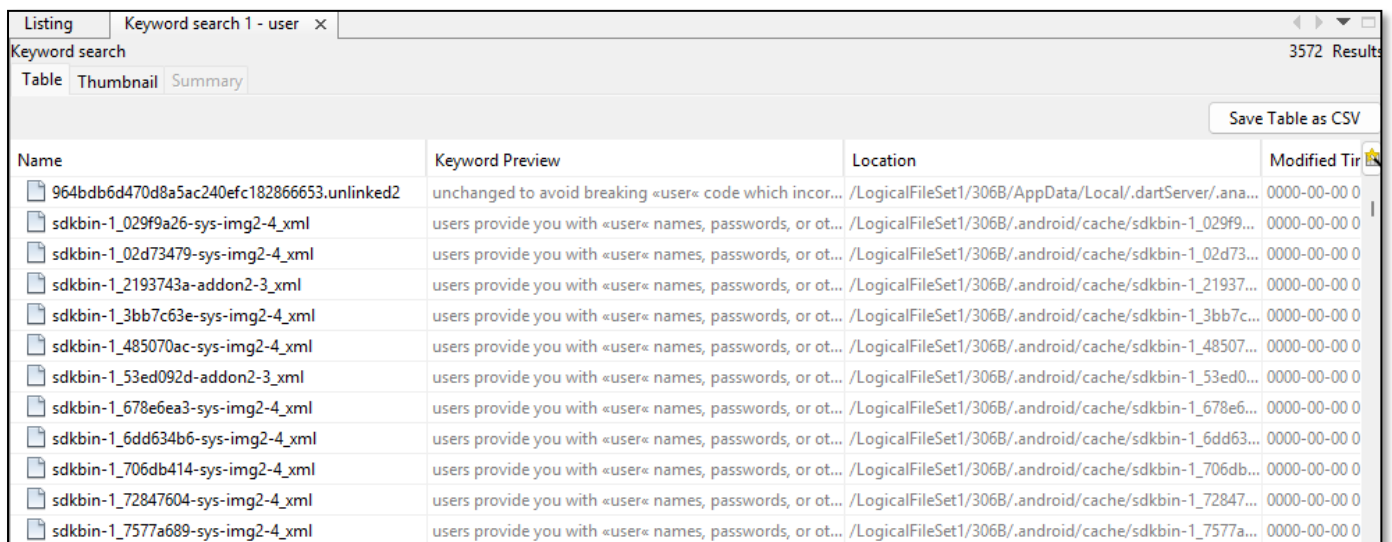


	Name	Date modified	Type	Size
Home	Cache	17-08-2026 14:34	File folder	
Gallery	Config	17-08-2026 14:50	File folder	
OneDrive	Export	17-08-2026 14:34	File folder	
	Log	17-08-2026 14:34	File folder	
Desktop	ModuleOutput	17-08-2026 14:50	File folder	
Downloads	Reports	17-08-2026 14:34	File folder	
Documents	.lock_autopsy	17-08-2026 14:34	Data Base File	1 KB
Pictures	autopsy	17-08-2026 14:50	Data Base File	496 KB
Music	f0000000_Microsoft_Word_EHF_EXP_1_1	17-08-2026 14:51	Microsoft Edge P...	2,914 KB

Observation:

Recovered 1 deleted file named **f0000000_Microsoft_Word_EHF_EXP_1_1.pdf** using the carving process. The file was identified as a **PDF**, marked as **Unallocated**, with a size of **2,983,912 bytes**, and was successfully extracted to the report output folder.

6. Keyword search results.



Name	Keyword Preview	Location	Modified Time
964bdb6d470d8a5ac240efc182866653.unlinked2	unchanged to avoid breaking «user» code which incor...	/LogicalFileSet1/306B/AppData/Local/.dartServer/ana...	0000-00-00 0
sdkbin-1_029f9a26-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_029f9...	0000-00-00 0
sdkbin-1_02d73479-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_02d73...	0000-00-00 0
sdkbin-1_2193743a-addon2-3.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_21937...	0000-00-00 0
sdkbin-1_3bb7c63e-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_3bb7c...	0000-00-00 0
sdkbin-1_485070ac-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_48507...	0000-00-00 0
sdkbin-1_53ed092d-addon2-3.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_53ed0...	0000-00-00 0
sdkbin-1_678e6ea3-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_678e6...	0000-00-00 0
sdkbin-1_6dd634b6-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_6dd63...	0000-00-00 0
sdkbin-1_706db414-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_706db...	0000-00-00 0
sdkbin-1_72847604-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_72847...	0000-00-00 0
sdkbin-1_7577a689-sys-img2-4.xml	users provide you with «user» names, passwords, or ot...	/LogicalFileSet1/306B/.android/cache/sdkbin-1_7577a...	0000-00-00 0

Observation:

Executed a search for the keyword **"user"**, which yielded **3,572 matching results**. Displayed the matching file names, keyword previews, locations, and other details in a tabular format.

7. Web activity analysis.

Web History								431 Results
Table	Thumbnail	Summary						
Page: 1 of 1		Pages: < >	Go to Page:		Save Table as CSV			
Source Name	S	C	O	URL	Date Accessed	Referrer URL	Title	Program
History			0	https://drive.google.com/?authuser=0	2026-04-28 10:58:55 IST	https://drive.google.com/?authuser=0	Google Drive	Google (
History			0	https://accounts.google.com/ServiceLogin?service=w...	2026-04-28 10:58:55 IST	https://accounts.google.com/ServiceLogin?service=w...	Google Drive	Google (
History			0	https://drive.google.com/accounts/SetOSID?authuser...	2026-04-28 10:58:55 IST	https://drive.google.com/accounts/SetOSID?authuser...	Google Drive	Google (
History			0	https://drive.google.com/?authuser=0&pli=1	2026-04-28 10:58:55 IST	https://drive.google.com/?authuser=0&pli=1	Google Drive	Google (
History			0	https://drive.google.com/drive/?pli=1	2026-04-28 10:58:55 IST	https://drive.google.com/drive/?pli=1	Google Drive	Google (
History			0	https://drive.google.com/drive/	2026-04-28 10:58:56 IST	https://drive.google.com/drive/	Google Drive	Google (
History			0	https://drive.google.com/drive/home	2026-04-28 10:59:01 IST	https://drive.google.com/drive/home	Home - Google Drive	Google (
History			0	https://drive.google.com/drive/home	2026-04-28 10:59:01 IST	https://drive.google.com/drive/home	Home - Google Drive	Google (
History			0	https://drive.google.com/drive/home	2026-04-28 10:59:01 IST	https://drive.google.com/drive/home	Home - Google Drive	Google (
History			0	https://www.google.com/search?q=chatgpt&eq=ch...	2026-04-28 11:21:57 IST	https://www.google.com/search?q=chatgpt&eq=ch...	chatgpt - Google Search	Google (
History			0	https://www.google.com/search?q=chatgpt&eq=ch...	2026-04-28 11:21:58 IST	https://www.google.com/search?q=chatgpt&eq=ch...	chatgpt - Google Search	Google (
History			0	https://www.google.com/acik?sa=L&pf=1&ai=DChsS...	2026-04-28 11:21:59 IST	https://www.google.com/acik?sa=L&pf=1&ai=DChsS...	ChatGPT	Google (
History			0	https://chatgpt.com/?utm_source=google&utm_med...	2026-04-28 11:21:59 IST	https://chatgpt.com/?utm_source=google&utm_med...	ChatGPT	Google (
History			0	https://www.google.com/search?q=chatgpt&eq=8g...	2026-04-28 11:22:39 IST	https://www.google.com/search?q=chatgpt&eq=8g...	chatgpt - Google Search	Google (
History			0	https://www.google.com/acik?sa=L&pf=1&ai=DChsS...	2026-04-28 11:22:44 IST	https://www.google.com/acik?sa=L&pf=1&ai=DChsS...	ChatGPT	Google (
History			0	https://chatgpt.com/?utm_source=google&utm_med...	2026-04-28 11:22:44 IST	https://chatgpt.com/?utm_source=google&utm_med...	ChatGPT	Google (
History			0	https://www.google.com/search?q=restaurant+image...	2026-04-28 11:29:00 IST	https://www.google.com/search?q=restaurant+image...	restaurant images in png - Google Search	Google (
History			0	https://www.google.com/search?q=restaurant+image...	2026-04-28 11:28:33 IST	https://www.google.com/search?q=restaurant+image...	restaurant images in png - Google Search	Google (
History			0	https://www.rawpixel.com/search/restaurant%20interi...	2026-04-28 11:28:38 IST	https://www.rawpixel.com/search/restaurant%20interi...	Restaurant Interior Images Free Photos, PNG Stickers,...	Google (
History			0	https://www.rawpixel.com/search/restaurant%20interi...	2026-04-28 11:28:39 IST	https://www.rawpixel.com/search/restaurant%20interi...	Restaurant Interior Images Free Photos, PNG Stickers,...	Google (

Observation:

Extracted **101 records** under **Web History** containing accessed URLs and timestamps. The results included browsing activity related to **Google Drive**, **Google Search**, **ChatGPT**, and **other websites**.

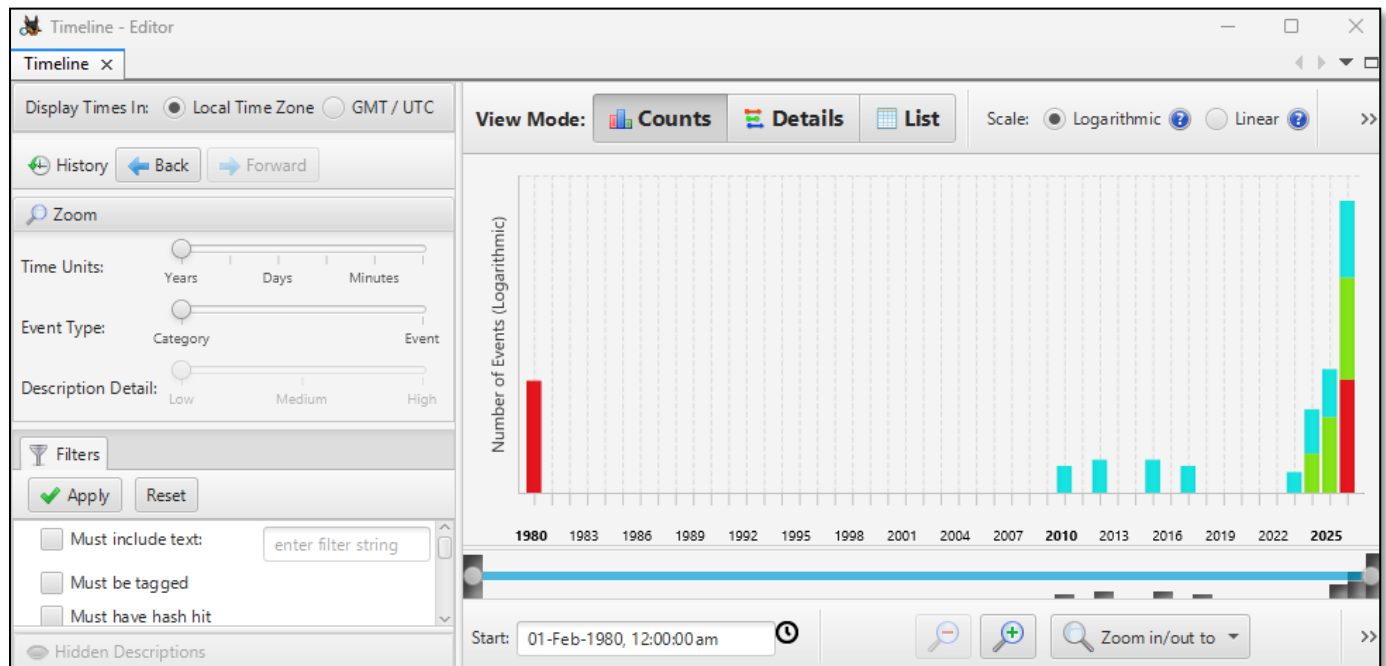
8. Metadata analysis of a selected file.

Metadata										19 Results
Table	Thumbnail	Summary								
Page: 1 of 1		Pages: < >	Go to Page:		Save Table as CSV					
Source Name	S	C	O	Date Modified	Program Name	Owner	Date Created	Data Source	Description	
</> aT5jUvT_qhlogs.doc				2024-08-02 05:25:00 IST	Microsoft Office Word	quick heal	2010-05-06 13:11:00 IST	LogicalFileSet1		
</> lqhlogs.doc				2015-02-20 15:25:00 IST	Microsoft Office Word	quick heal	2015-02-20 15:25:00 IST	LogicalFileSet1		
</> lqhlogs.doc				2015-02-20 15:25:00 IST	Microsoft Office Word	quick heal	2015-02-20 15:25:00 IST	LogicalFileSet1		
</> zT2xilN_qhlogs.xls				2024-07-25 04:37:20 IST				LogicalFileSet1		
</> aT5jUvT_qhlogs.doc				2024-08-02 05:25:00 IST	Microsoft Office Word	quick heal	2010-05-06 13:11:00 IST	LogicalFileSet1		
</> EHF_Experiment1.docx.pdf								LogicalFileSet1	EHF_Experiment1.docx	
</> Experiment4.docx.pdf								LogicalFileSet1	Experiment4.docx	
</> Experiment 3.docx.pdf								LogicalFileSet1	Experiment 3.docx	
</> zT2xilN_qhlogs.xls				2024-07-25 04:37:20 IST				LogicalFileSet1		
</> EHF_Experiment1.docx				2026-07-16 07:28:00 IST	Microsoft Office Word	306B	2026-07-16 07:28:00 IST	LogicalFileSet1		
</> EHF_Experiment2.docx.pdf								LogicalFileSet1	EHF_Experiment2.docx	
</> EHF_EXP5.docx				2026-08-12 01:22:00 IST	Microsoft Office Word	Radha Ursekar	2026-08-11 13:32:00 IST	LogicalFileSet1		
</> EHF_Experiment5.docx.pdf								LogicalFileSet1	EHF_Experiment5.docx	
</> Web Lab_exp1.pdf				2026-07-13 04:17:20 IST			2026-07-13 04:17:20 IST	LogicalFileSet1	Microsoft Word - Lab-Manual-web-lab-2026-27_witl	
</> Exp-2.pdf								LogicalFileSet1	Exp-2	
</> QlikView Reference Manual.pdf				2012-12-06 16:40:54 IST		QlikView	2012-12-06 16:40:54 IST	LogicalFileSet1	QlikView Reference Manual	
</> Text Sidebar (Annual Report Red and Black design).				2012-03-05 20:27:00 IST	Microsoft Office Word			LogicalFileSet1	Text Sidebar (Annual Report Red and Black design)	
</> Text Sidebar (Annual Report Red and Black design).				2012-03-05 20:27:00 IST	Microsoft Office Word			LogicalFileSet1	Text Sidebar (Annual Report Red and Black design)	
Data Content										
Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences	
Result: 1 of 1		Result: < >	Metadata							
Type	Value								Source(s)	
Owner	quick heal								org.sleuthkit.autopsy.keywordsearch	
Date Created	2010-05-06 13:11:00 IST								org.sleuthkit.autopsy.keywordsearch	
Source File Path	/LogicalFileSet1/Users/306B/Downloads/aT5jUvT_qhlogs.doc									
Artifact ID	-9223372036854775669									

Observation:

Inspected metadata results showing details such as **file name, date modified, program name, owner, date created, and data source**. Verified metadata information for the selected file within **LogicalFileSet1**.

9. Timeline analysis showing user activities.

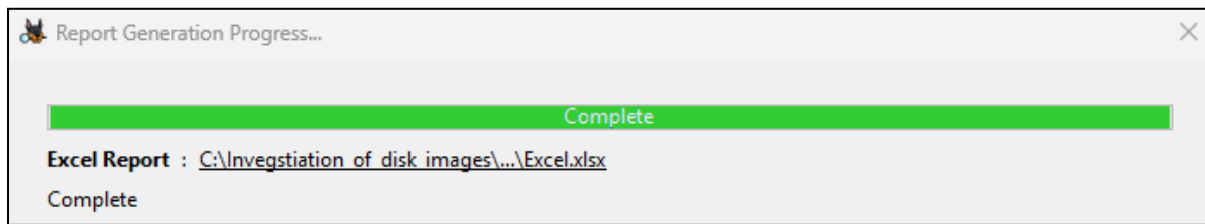


Observation:

Displayed chronological user and system events on a **logarithmic timeline**. The timeline showed activity across different years, with a higher concentration of events around **2025–2026**.

10. Generated forensic investigation report.

The screenshot shows the 'Select and Configure Report Modules' dialog box. It has a title bar and a main area divided into two sections. The left section, titled 'Report Modules:', contains a list of modules with radio buttons: HTML Report, Excel Report (selected), Files - Text, Data Source Summary Report, Save Tagged Hashes, Extract Unique Words, TSK Body File, Google Earth KML, CASE-UCO, and Portable Case. The right section contains a description: 'A report about results and tagged items in Excel (XLS) format.' Below this, there is a text box with the message: 'This report will be configured on the next screen.'



	A	B	C	D	E	F	G	H	I	J
1	Summary									
2										
3	Case Name:	Invegstiation_of_disk_images								
4	Case Number:	3663								
5	Number of data sources in cas	1								
6	Case Notes:	None								
7	Examiner:	Shivam Pimple								
8										
9										
10										
11										
12										
13										
14										
15										
16										
17										
18										
19										
20										
21										
22										
23										
24										
25										
26										
27										
	< >	Summary	Operating System Information	Chromium Extensions	Web Bookmarks	Web Account Type	Web Cook			

Observation:

Selected the **Excel Report** module to export the investigation results. Successfully generated the report containing the **case summary**, **case number 3663**, **one data source**, **case notes**, and **examiner Shivam Pimple**.